

Prepared By	09-Jun-2012	Mr.Chatchawan S.	 RENEWCOILS ENGINEER & SUPPLY CO.,LTD. Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnauy W.		
Certified By	09-Jun-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-003-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-006**



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 -9978

Welding Procedure Specification (WPS)

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**
WPS No. **RNC-WPS-006** Date **9-Jun-12** Revision No. **0**
Supporting PQR No.(s) **RNC-PQR-006** Date **9-Jun-12** Type(s) **Manual**
Welding Process(es) **Gas Tungsten Arc Welding + Shield Metal Arc welding**
(GTAW + SMAW)

JOINT (QW-402)

Joint Design **Single V**
Backing (Yes) ☐ (No) ☒ X
Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes _____

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **1** Group No. **1.2** to P-No. **1** Group No. **1.2**
Material Specification Type and Grade **SA 333 Gr.1.6 SA420 WPL6** TO **SA 333 Gr.1.6 SA420WPL6**
Chem. Analysis and Mech. Prop. _____
Thickness Range :
Base Metal : Groove **1.6 mm - 14.22 mm.** Fillet **All**
Pipe Dia. Range : Groove **All** Fillet **All**
Other _____

FILLER METALS (QW-404)

	GTAW	SMAW
Spec. No. (SFA)	5.28	5.5
AWS No. (Class)	ER-70S-G	E8016-G
F-No.	6	4
A-No.	1	1
Size of Filler Metals	Ø 2.0 & 2.4 mm	Ø 3.2 mm.
Deposited Weld Metals	2.0 mm	5.11 mm.
Thickness Range :		
Groove	4 mm Max.	10.22 mm. Max.
Fillet	All	All
Electrode-Flux (Class)	-	-
Flux Trade Name	-	-
Consumable Insert	-	-
Other	Kobelco TGS- 1N or Equivalents	Kobelco NB- 1 or Equivalents

POSITION (QW-405)

Position(s) of Groove **ALL**
Welding Progression Up ☒ X Down ☐
Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **593 - 649°C**
Time Range **Holding Time 1 Hour**
Notes **Heating & Cooling Rate = 200 ° Max.**

PREHEAT (QW-406)

Preheat Temp. Min. **10°C**
Interpass Temp. Max. **238°C**
Preheat Maintenance **N/A**
Notes _____

GAS (QW-408)

	Percent Composition		
	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	13-17Ltr/Min
Trailing	-	-	-
Backing	-	-	-

WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. RNC-WPS-006 Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☒ AC ☒ DC Polarity EN(GTAW)+ EP(SMAW)
 Amps (Range) 90-140(GTAW) + 80-130(SMAW) Volts (Range) 9-13(GTAW) + 18-24(SMAW)
 Tungsten Electrode Size and Type Ø 2.4 mm , Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size 8.0 mm - 19.0 mm
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) 6.0-10.0 Cm/Minute
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER 70S-G	2.4	DC EN	90-130	9-12	6.0-10.0	
2nd	GTAW	ER 70S-G	2.4	DC EN	100-140	10-13	6.0-10.0	
3 & Over	SMAW	E8016-G	3.2	AC/DCEP	80-130	18-24	6.0-10.0	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of ASME-LX, 2007 latest Revision.

Prepared by: Mr. Chatchanan S. Signed: [Signature] Date: 9/06/12
 Reviewed by: Mr. Saksan D. Signed: [Signature] Date: 9-06-12
 Approved by: AMNAY W. Signed: [Signature] Date: 29-06-12



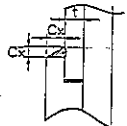
Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

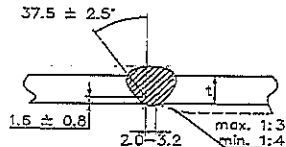
Procedure Qualification Record (PQR)

ATTACHED FIGURE JOINT SKETCH



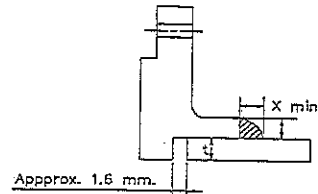
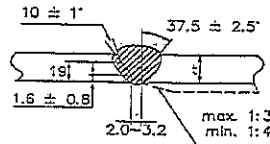
C_x (min.) = $1.25t$ but not less than 3.2 mm.

JOINT No. A1



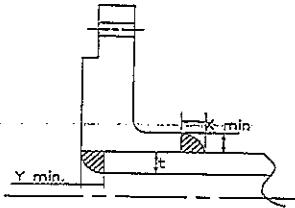
Misalignment (max.) = 1.5 mm.

JOINT No. B1 & B2



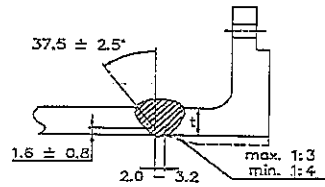
X min. = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



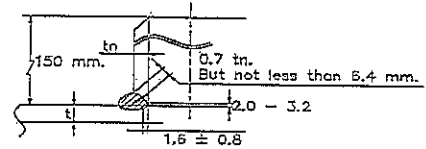
X min. = The lesser of $1.4t$ or thickness of the hub.
Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



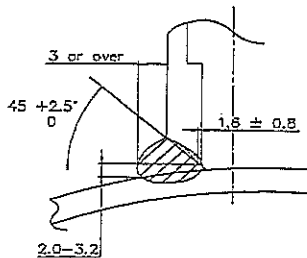
Misalignment (max.) = 1.5 mm.

JOINT No. B3

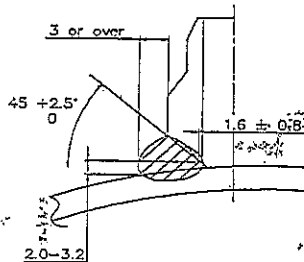


Misalignment (max.) = The lesser of 3.2 mm. or $0.5t$.

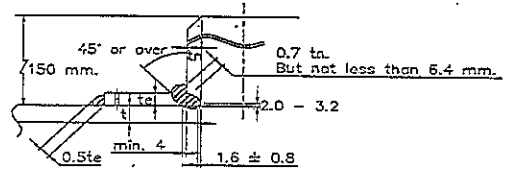
JOINT No. B4



JOINT No. C1

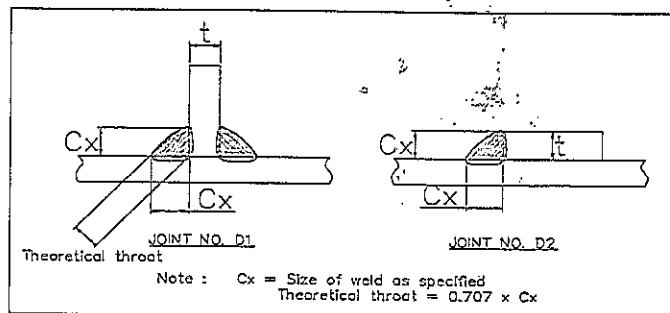


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or $0.5t$.

JOINT No. B5



Procedure Qualification Record (PQR)

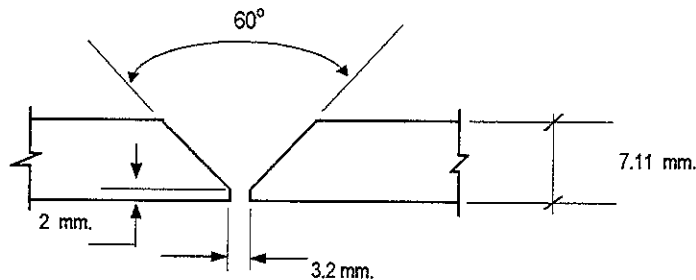
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-006

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-006	Date :	13 June 2012
WPS Number :	RNC-WPS-006		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)					
Material Spec.	A333	To	A333	Temperature	610 °C				
Type or Grade	Gr.6	To	Gr.6	Hold Time	1 HR.				
P. No.	P.1	To	P.1	Others					
Thickness of Test Coupon		7.11 mm.		GAS (QW-408)					
Diameter of Test Coupon		Ø 6"							
Others									
FILLER METALS (QW-404)				Percent Composition					
					Gas (es)	Mixture	Flow Rate		
				Shielding	Argon	99.95%	15 L/Min.		
				Trailing	-	-	-		
				Backing	-	-	-		
SFA Specification				ELECTRICAL CHARACTERISTICS (QW-409)					
AWS Classification		GTAW	A 5.28	SMAW		A 5.5	Current	DC (GTAW)	DC (SMAW)
Filler Metal F-No.		6	4	Polarity		EN	EP		
Weld Metal Analysis A-No.		1	1	Amperage		92-102 for GTAW and 85-94 for SMAW	Voltage		10 for GTAW 22 for SMAW
Size of Filler Metal		Ø 2.4 mm.	Ø 3.2 mm.	Tungsten Electrode Size		Ø 2.4 mm. 2% thoriated.EWTh-2			
Others		Kobelco TGS-1N	Kobelco NB-1	Others		N/A			
Weld Metal Thickness		-		TECHNIQUE (QW-410)					
POSITION (QW-405)				Travel Speed		7 - 9 cm/min.			
Position of Groove		6G		String or Weave Bead		Both			
Progression of Welding (Uphill or Down)		Up Hill		Oscillation		N/A			
Others		N/A		Multipass or single Pass		Multiple Pass			
PREHEAT (QW-406)				Multipass or Single Electrodes		Multiple Pass			
Preheat Temperature		Ambient Temperature		Others					
Interpass Temperature (max.)		220° C				N/A			
Others		N/A							

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- RNC-PQR-006

Tensile Test Report No.TS-12-06-104

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
2365/12TR1	19.38	6.84	132.559	60.252	454.529	OUT OF WELD
2365/12 TR2	19.44	6.81	132.386	59.664	450.678	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-12-06-102

Type and Figure Number	Type of Bend	Indication	Results
2365/12 BF1	Transverse face bend	No Open Discontinuity	Accepted
2365/12 BF2	Transverse face bend	No Open Discontinuity	Accepted
2365/12 BR1	Transverse root bend	No Open Discontinuity	Accepted
2365/12 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

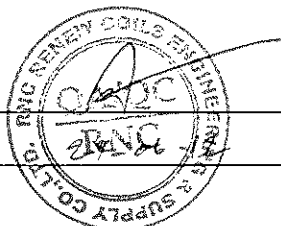
Type of Test Radiographic examination results Accepted Report No.PQR - 004
Hardness Testing Report No. HN-12-06-079
Macrostructure Report No. MA-12-06-101
Impact Testing Report No.IM-12-06-138

Welders Name Mr.Niwat R. Welder Number -
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date :



Inspected by :

Certified by :

Date :

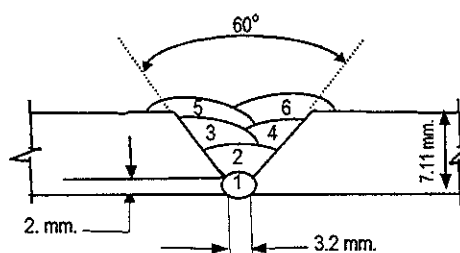
Mr.Nathawut M.

Mr.Chaloemket R.

06/06/2012

PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : RNC-PQR-006
		WPS No. : RNC-WPS-006
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST
		CONDITION AND RESULT

Welding Sequences for RNC-PQR-006 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or	Wire Feed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Weave Bead	Speed				
1	GTAW	ER70S-G	2.4	99	10	DCEN	-	-	8	7.42	15	-
2	GTAW	ER70S-G	2.4	102	10	DCEN	-	-	9	6.80	15	-
3	SMAW	E8016-G	3.2	94	22	DCEP	-	-	7	17.72	-	-
4	SMAW	E8016-G	3.2	85	22	DCEP	-	-	7	16.02	-	-
5	SMAW	E8016-G	3.2	85	22	DCEP	-	-	7	16.02	-	-
6	SMAW	E8016-G	3.2	87	22	DCEP	-	-	8	14.35	-	-



PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250
Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

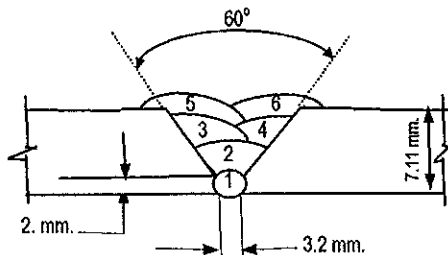
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : PQR		Location : RNC Work Shop		Report No. PQR-004	
Customer Order No. : N/A		WPS No. : RNC-WPS-006		Date : 13-Jun-12			

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.NIWAT R.	GTAW+SMAW	6G	ASME IX	A333Gr.6	ø 6" Thk. 7.11 mm.	ER70S-G/E8016-G	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	DCEN	99	10	8	117	7.42	Argon	15	-	-
2	GTAW	ER70S-G	2.4	DCEN	102	10	9	126	6.80	Argon	15	-	-
3	SMAW	E8016-G	3.2	DCEP	94	22	7	180	17.72	-	-	-	-
4	SMAW	E8016-G	3.2	DCEP	85	22	7	200	16.02	-	-	-	-
5	SMAW	E8016-G	3.2	DCEP	85	22	7	207	16.02	-	-	-	-
6	SMAW	E8016-G	3.2	DCEP	87	22	8	220	14.35	-	-	-	-

PAE Inspection By

Approved By

SIGNATURE :

NAME :

DATE :

PAE
Mr. Natthawut M.
14-Jun-2012

SIGNATURE :

NAME :

DATE :

Witness By
RNC
Chatchawan S.
14-Jun-2012

SIGNATURE :

NAME :

DATE :

ANUSAY W.
29-Jun-12



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36. T. TABMA. A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 -9978

Radiographic Examination Report

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Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Mechanical Test Report



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnplm.co.th



21712

TESTING
No. 010

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-12-06-104

Revision No. 0

Test Product	PQR No. RNC-PQR-006	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A333 Gr.6	Type of Test	Reduced Section Tension
Dimension	Ø 6" Sch 40 (7.11 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	19 June 2012	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	21 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	2365/12 TR1	N/A	6.84	19.38	132.559	N/A	N/A	N/A	60.252	454.529	N/A	OUT OF WELD
2	2365/12 TR2	N/A	6.81	19.44	132.386	N/A	N/A	N/A	59.664	450.678	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-006

Welding Process : GTAW+SMAW

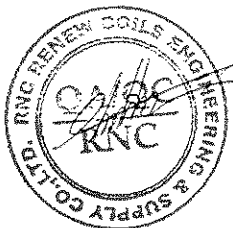
Filler Metal/Electrode Spec't : ER-70S-G+E8016-G/TGS-1N+NB 1

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmpm.co.th

21720

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-12-06-102

Revision No. 0

Test Product	PQR No. RNC-PQR-006	Material Characterization	Pipe
Project Name	N/A	Test Method	ASME Sec. IX : 2010
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A333 Gr.6	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 6" Sch 40 (7.11 mm Thk.)	Diameter of Mandrel	30 mm
Date Received	19 June 2012	Bend Angle	180 Degree.
Date Tested	21 June 2012	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	2365/12 BF1	Transverse face bend	7.20x38.08x161.74 mm	No Open Discontinuity	N/A	-
2	2365/12 BF2	Transverse face bend	7.33x38.09x163.47 mm	No Open Discontinuity	N/A	-
3	2365/12 BR1	Transverse root bend	7.27x38.16x162.85 mm	No Open Discontinuity	N/A	-
4	2365/12 BR2	Transverse root bend	7.28x38.18x159.98 mm	No Open Discontinuity	N/A	-

Additional details : WPS No. RNC-WPS-006

Welding Process : GTAW+SMAW

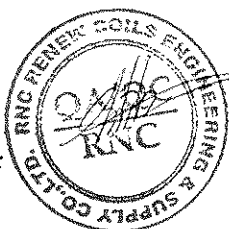
Filler Metal/Electrode Spec't : ER-70S-G+E8016-G/TGS-1N+NB 1

Reported by : 

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12

Approved by : 

(Mr.Pavarat Preedawiphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnplm.co.th



21728

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-12-06-138

Revision No. 0

Test Product	PQR No. RNC-PQR-006	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ⁰¹
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A333 Gr.6	Type of Test	Charpy V-Notch Impact
Dimension	Ø 6" Sch 40 (7.11 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	19 June 2012	Equipment 's Capacity	542 Joules
Date Tested	22 June 2012	Ambient Temperature	23 + 5 °C

Item No.	Test No.	Test Temperature (°C)	Size of Specimen			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	2365/12-1	-20°C	5.014	10.015	54.96	133.32	126.73	Base Metal	-
2	2365/12-2	-20°C	5.011	10.015	54.94	136.60			
3	2365/12-3	-20°C	5.014	10.009	54.86	110.28			
4	2365/12-1	-20°C	5.016	10.015	54.95	111.94	91.31	Weld Metal	-
5	2365/12-2	-20°C	5.012	10.011	54.96	85.70			
6	2365/12-3	-20°C	5.014	10.014	54.95	76.28			
7	2365/12-1	-20°C	5.017	10.013	54.95	153.27	126.39	HAZ	-
8	2365/12-2	-20°C	5.008	10.013	54.97	114.56			
9	2365/12-3	-20°C	5.012	10.009	54.93	111.35			

Additional details : WPS No. RNC-WPS-006

Welding Process : GTAW+SMAW

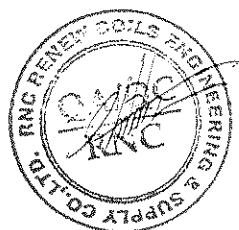
Filler Metal/Electrode Spec't : ER-70S-G+E8016-G/TGS-1N+NB 1

Reported by :

(Mr. Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12



Approved by :

(Mr. Pavaret Freedawiphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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LPN PLATE MILL

Macrostructure Analysis Report

Page 1 of 1

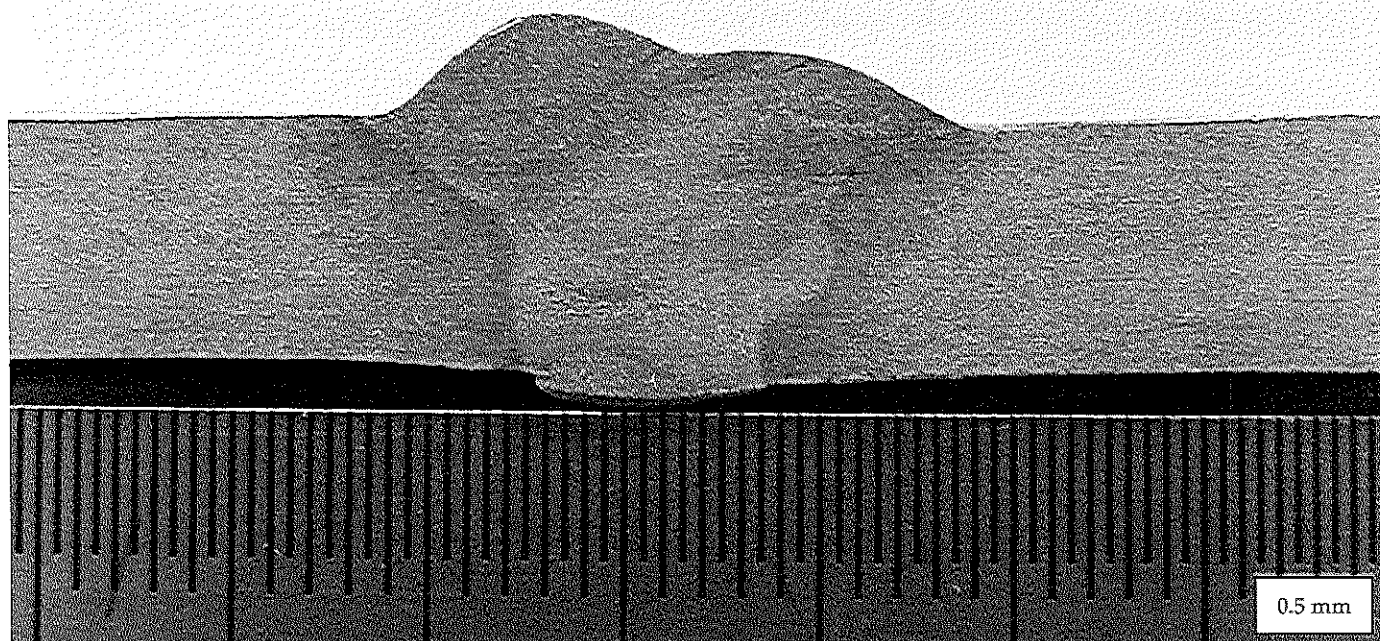
Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. MA-12-06-101

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-006	Material Characterization	Pipe
Test No.	2365/12 MA	Test Method	ASME IX : 2010
Project Name	N/A	Standard of Test	ASME IX : 2010
Project No.	N/A	Etchant Reagent	75ml H ₂ O+25ml HNO ₃ (65%)
Material Specification	A333Gr.6	Temperature	28 °C
Dimension	Ø6"Sch40(7.11 mm Thk.)	Time of Etching	5 minutes
Welding Process/Position	GTAW+SMAW	Surface Preparation	Polishing with sand paper
Type of Joint	Butt Joint		240, 400, 600, 800, 1000
Date Received	19 June 2012	Date Tested	21 June 2012



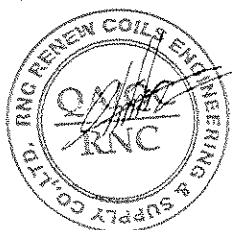
Additional details : The macrostructure shows the complete fusion.Weld metal and heat affected zone were free of cracks
 Filler/Electrode Spec't : ER70S-G+E8016-G TGS-IN+NB 1

Reported by : N. D. Saeng

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date : 29/06/12



Approved by : N. P. Pree

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date : 29/06/12

LPN PLATE MILL

Notes

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited 12406

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. HN-12-06-079

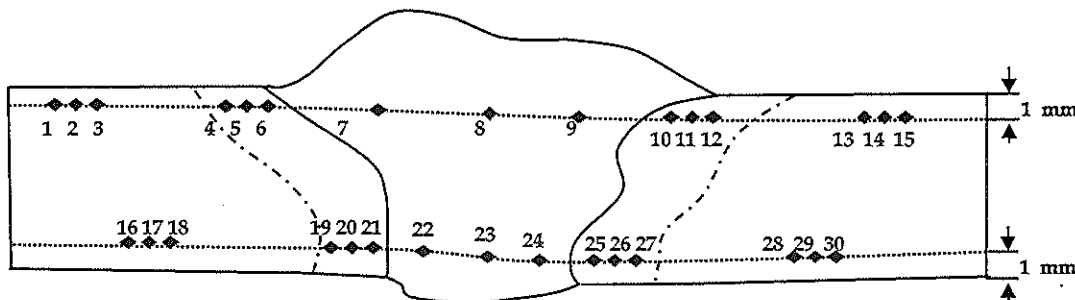
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-006	Material Characterization	Pipe
Test No.	2365/12 HN	Test Method	ASTM E92-82 : 2003
Project Name	N/A	Standard of Test	ASTM Sec. IX : 2010
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A333Gr.6	Equipment	Mitutoyo, HV-115
Dimension	Ø6"Sch40(7.11 mm Thk.)	Indentor Size	Diamond, pyramid 136 deg.
Welding Process/Position	GTAW+SMAW	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	19 June 2012	Cal. Block	No. 62137 No. A62072
Date Tested	21 June 2012	Ambient Temperature	25.8 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale)
1	BASE	129.9	16	BASE	128.7	*****		
2	BASE	131.3	17	BASE	127.9			
3	BASE	130.8	18	BASE	128.8			
4	HAZ	138.0	19	HAZ	141.1			
5	HAZ	148.4	20	HAZ	147.5			
6	HAZ	192.6	21	HAZ	164.6			
7	WELD	218.3	22	WELD	170.5			
8	WELD	185.3	23	WELD	165.4			
9	WELD	202.2	24	WELD	160.1			
10	HAZ	172.6	25	HAZ	154.2			
11	HAZ	149.4	26	HAZ	147.1			
12	HAZ	136.8	27	HAZ	140.9			
13	BASE	130.1	28	BASE	129.1			
14	BASE	129.6	29	BASE	128.7			
15	BASE	129.2	30	BASE	128.2			

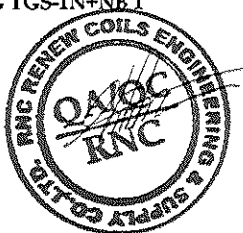
Additional details: Filler/Electrode Spec't: ER70S-G+E8016-G TGS-1N+NB1

Reported by: N. P. D.

(Mr. Keerati Sa-ngoen)

Laboratory Section Manager

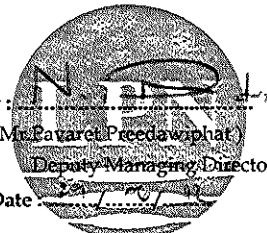
Date: 29/06/12

Approved by: N. P. D.

(Mr. Pavaree Freedawiphat)

Deputy Managing Director

Date: 29/06/12



LPN PLATE MILL

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Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

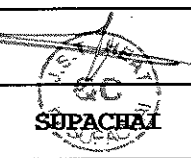
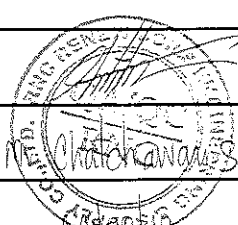
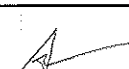
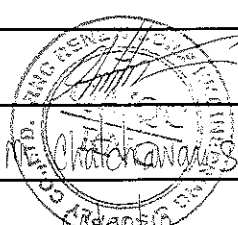
TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

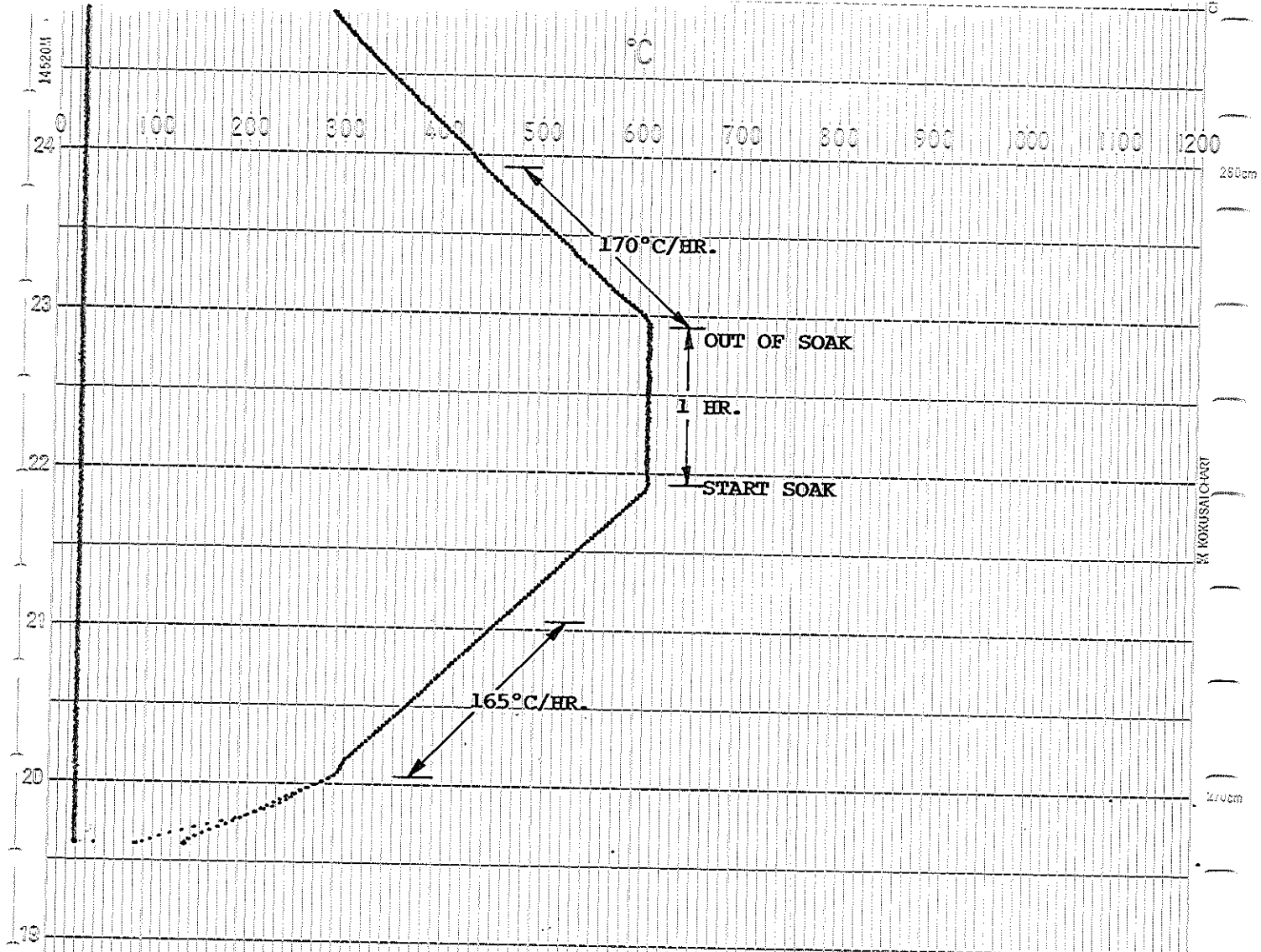
Post weld Heat Treatment Report

HEAT		J.S.T. HEAT TREATMENT COMPANY LIMITED	
POST WELD HEAT TREATMENT REPORT			
CLIENT : RENEW COILS ENGINEERING & SUPPLY CO.-LTD.			
CONTRACTOR :	J.S.T. HEAT TREATMENT CO.,LTD.		PAGE : 1
PROJECT : PQR			
P.W.H.T.METHOD : ☒ LOCAL ☐ FURNACE			
COMPLETION DATE : 14-06-2012			REPORT NO.PW-001
CHART SPEED : ☒ 25 MM/Hr, ☐ 50 MM/Hr, ☐ 100 MM/Hr.			
RECORDER NO : EH-066A042		MATERIAL : A335 GR.6,A106 GR.B	
LOCATION : ☒ SHOP JST. ☐ FIELD		UNIT :	

HEAT TREATMENT CONDITION/RESULTS						
DESCRIPTION	LOADING TEMP(C°)	HEATING RATE (C°/Hr)	HOLDING TEMP (MIN-MAX)	HOLDING TIME (Hr)	COOLING RATE (C°/Hr)	UNLOADING TEMP (C°)
REQUEST	300	200	593-649	1 HR.	200	300
ACTUAL	300	165	610	1 HR.	170	300

DWG NO.,LINE NO.&CLASS,SHT.NO.SPOOL	WELD NO.	SIZE (")	THK. (MM)	WELD TYPE	T/C NO.
RNC-PQR-005		2"	3.91	BW	5
RNC-PQR-006		6"	7.11	BW	3,4
RNC-PQR-009		6"	10.97	BW	1,2
TOTAL	3	JOINTS	TOTAL	14	D.B.

CHECKED BY J.S.T. HEAT TREATMENT CO.,LTD.		REVIEWED BY	APPROVED BY	APPROVED BY
SIGN :				
NAME :	SUPACHAI K.		ANNUAY.W.	
DATE :	14-06-2012		29-06-12	



DATE. 14-06-2012

RECORDER NO. EH - 066A042

CHART NO. PW-001

CHART SPEED. 25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO., LTD.

PROJECT. PQR

DWG NO. RNC-PQR-005

SIZE. 2" THK. 3.91mm.

DWG NO. RNC-PQR-006

SIZE. 6" THK. 7.11mm.

DWG NO. RNC-PQR-009

SIZE. 6" THK. 10.97mm.

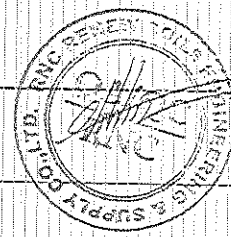
MATERIAL. A333 GR.6, A106 GR.B

START TIME. 21.30 PM.

TECHNICIAN. MR. ANU S.

TC COLOUR

1. RED
2. BLACK
3. SKY BLUE
4. GREEN
5. BROWN



COOPERHEAT

Certificate of Calibration / Test



INNOVATORS IN THERMAL TECHNOLOGY

Certificate No.

JST-053/2012

Date of Issue

1/2/2012

Signatory (Print)

Supachai K

Equipment Description

RECORDER "CHINO" EH SERIES

Input or Sensor Type

K

Output or Range

0-1200°C

Cooperheat Identification

Serial No.

Next Calibration Due

EH-066A042

1/8/2012

Customer Name

Customer Order No.

Works Order No.

Test Parameter	200	400	600	800	1000	1200
Prior to Test	200	400	600	800	1000	1200
Post Test	200	400	600	800	1000	1200

Instrument Used	Model	Model	Date Calibrated
FLUKE	714	6986010	25 June 2011

All measurement are performed with equipment that is traceable to National Standard unless otherwise

Signature

Date

1/2/2012

Customer

J.S.T. HEAT TREATMENT CO., LTD.

Certificate of Conformance

This material has been tested, inspected and approved by Cooperheat quality control in compliance with Cooperheat's BS EN ISO 9001 : 2000 Quality Management System.

43000 (100 mtrs)	TYPE "K" (NiCr-NiAl) Twin Thermocouple wire conforms to:- BS EN 60584-1: BS EN 60584-2: ANSI-MC96-1: Temperature Measurement Thermocouples. 1982
------------------	---

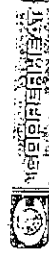
Batch No. 716718121162

QC Inspector's Sig:

[Signature]

Date

15/4/2008



CERTIFICATE OF CALIBRATION

mmc
(THAILAND) LTD

Issued By Marske Machine Hire & Calibration Centre

Date of Issue 25 June 2011

Certificate Number

E10544

Page 1 of 3 Pages



CALIBRATION
No. 0035

Marske Machine (Thailand) Co., Ltd
2nd Floor, The Seaboard Group Building
2/58 Highway 3191, MapTaPhut, Muang, Rayong 21150
Tel : (038) 682992-3, (038) 691563-4 Fax : (038) 682988
Email : sales@mmc-thai.com

Approved Signatory

[Signature]
Marut L.

Customer : J.S.T. Heat Treatment Co., Ltd.
29/9 Soi Charoenmitr (Ekamai 10), Sukhumvit 63 Rd., Klongton Nue
Wattana, Bangkok 10110

Date Received : 24 June 2011

Instrument -	System ID :	EC00745	Job Number :	0000113262
	Description :	Thermocouple Calibrator		
	Manufacturer :	FLUKE		
	Model Number :	714		
	Serial Number :	6986010	Last Certificate Number :	E00544
	Procedure Version :	EP0008	Last Calibration Date :	15/06/2010

Environmental Conditions

Temperature : 23°C +/- 2°C
Relative Humidity : 55% +/- 15%

Mains Voltage : 220V +/- 10V
Mains Frequency : 50Hz +/- 1Hz

Comments

The instrument was placed in lab and allowed to stabilise before calibrate.
The direct comparison technique has been applied to calibrate.
All tests passed Mfg'Spec Model 714 Rev 1,10/97.

Traceability Information

Instrument description	Serial number	Certificate number	Cal. Date	Cal. Period
3041 Precision Multi-Product Calibrator	F1031B8	19721	26/04/2011	52
3458A Multimeter	MY45043338	EE-0052-11	27/05/2011	52

Calibrated By : Tassanee K.

[Signature]

Date of Calibration : 25 June 2011

This certificate provides traceability of measurement to recognised National Standards, and to the units of measurement realised at the National Physical Laboratory or other recognised National Standards laboratories.
Copyright of this certificate is owned by the issuing laboratory and may not be reproduced except with the prior written approval of the issuing laboratory.
This certificate complies with the requirements of BS EN ISO 10012:2003.

Quality Isn't Expensive It's Priceless

CERTIFICATE OF CALIBRATION

mmc
(THAILAND) LTD

Certificate Number
E10544

Page 2 of 3 Pages

Test Title	Tolerance	STD Value	UUC Reading	% Of Spec.	Uncertainties
------------	-----------	-----------	-------------	------------	---------------

Calibration Information.

The uncertainty limits quoted refer to the measured values only, with no account being taken of the instrument ability to maintain its calibration.

The reported expanded uncertainty of measurement is state as the standard uncertainty of measurement multiplied by a coverage factor $k=2$ corresponding to a coverage probability of approximately 95%.

Function : DC Voltage Measurement

Range 75 mVDC	19.8uV	10.000 0mV	10.000mV	0%	4uV
Range 75 mVDC	28.8uV	-10.0000mV	-9.990mV	35%	6uV
Range 75 mVDC	19.8uV	30.000 0mV	30.010mV	51%	4uV
Range 75 mVDC	19.8uV	50.000 0mV	50.010mV	51%	4uV
Range 75 mVDC	19.8uV	70.000 0mV	70.010mV	51%	4uV

Function : Temperature Measurement

1370 °C T/C Type K	0.83°C	-100.00°C	-99.7°C	36%	0.17°C
1370 °C T/C Type K	0.76°C	0.00°C	0.3°C	39%	0.16°C
1370 °C T/C Type K	0.74°C	100.00°C	100.2°C	27%	0.15°C
1370 °C T/C Type K	0.73°C	500.00°C	500.2°C	27%	0.15°C
1370 °C T/C Type K	0.76°C	1 000.00°C	1 000.3°C	39%	0.16°C
1200 °C T/C Type J	0.74°C	-100.00°C	-99.6°C	54%	0.15°C
1200 °C T/C Type J	0.70°C	0.00°C	0.3°C	43%	0.14°C
1200 °C T/C Type J	0.68°C	100.00°C	100.3°C	44%	0.14°C
1200 °C T/C Type J	0.68°C	500.00°C	500.3°C	44%	0.14°C
1200 °C T/C Type J	0.67°C	1 000.00°C	1 000.3°C	45%	0.14°C
1760 °C T/C Type R	2.5°C	0.00°C	1.0°C	40%	0.50°C
1760 °C T/C Type R	1.8°C	100.00°C	100.0°C	0%	0.36°C
1760 °C T/C Type R	1.4°C	500.00°C	500.0°C	0%	0.28°C
1760 °C T/C Type R	1.3°C	900.00°C	900.0°C	0%	0.26°C
1760 °C T/C Type R	1.2°C	1 300.00°C	1 300.0°C	0%	0.24°C

Function : DC Voltage Source

Range 75 mVDC	28.8uV	9.998mV	10.00mV	8%	6uV
Range 75 mVDC	28.8uV	29.998mV	30.00mV	8%	6uV
Range 75 mVDC	28.8uV	49.997mV	50.00mV	10%	6uV
Range 75 mVDC	28.8uV	69.997mV	70.00mV	10%	6uV

Function : Temperature Source

1370 °C T/C Type K	0.83°C	-100.27°C	-100.0°C	33%	0.17°C
1370 °C T/C Type K	0.76°C	-0.24°C	0.0°C	32%	0.16°C
1370 °C T/C Type K	0.74°C	99.86°C	100.0°C	19%	0.15°C
1370 °C T/C Type K	0.73°C	499.85°C	500.0°C	21%	0.15°C
1370 °C T/C Type K	0.76°C	999.88°C	1 000.0°C	16%	0.16°C
1200 °C T/C Type J	0.74°C	-99.98°C	-100.0°C	3%	0.15°C
1200 °C T/C Type J	0.70°C	0.07°C	0.0°C	10%	0.14°C
1200 °C T/C Type J	0.68°C	100.04°C	100.0°C	6%	0.14°C
1200 °C T/C Type J	0.68°C	500.04°C	500.0°C	6%	0.14°C
1200 °C T/C Type J	0.67°C	999.95°C	1 000.0°C	7%	0.14°C

CERTIFICATE OF CALIBRATION



Certificate Number
E10544

Page 3 of 3 Pages

Item Title	Tolerance	STD Value	UUC Reading	% Of Spec.	Uncertainty
0 °C T/C Type R	2.5°C	0.75°C	0.0°C	30%	0.50°C
50 °C T/C Type R	1.8°C	99.50°C	100.0°C	28%	0.36°C
100 °C T/C Type R	1.4°C	499.54°C	500.0°C	33%	0.28°C
500 °C T/C Type R	1.3°C	899.75°C	900.0°C	19%	0.26°C
1000 °C T/C Type R	1.2°C	1 299.84°C	1 300.0°C	13%	0.24°C

of calibration.



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Consumable Certificate

TKW110514 1008100

INSPECTION CERTIFICATE
(検査証明書)PURCHASER
需要家COVERED ELECTRODE
被覆アーク溶接棒CERTIFICATE No.
証明書番号: 120112100
DATE OF ISSUE
発行日: 2011-08-18

TRADE DESIGNATION 品名 (型番)		DIMENSION 寸法 (mm)		MFG.No 製造番号		APPLICABLE SPECIFICATION AND CLASSIFICATION 適用規格及び種類					
NB-1		(3.2) mm 1/8 inch		A171		AMS A5.5 E8016-6 ASME SFA-5.5 E8016-6					
CHEMICAL COMPOSITION 化学成分 (%) ACCORDING TO EN10204 TYPE 3.1											
ELEMENTS 成分	C	SI	MM	P	S	NI					
DEPOSITED METAL 溶着金属	0.05	0.62	1.04	0.008	0.003	1.69					
ELEMENTS 成分											
DEPOSITED METAL 溶着金属											
TENSILE PROPERTIES OF DEPOSITED METAL 溶着金属の引張性能						IMPACT PROPERTIES OF DEPOSITED METAL 溶着金属の衝撃性能			HARDNESS PROPERTIES 硬さ		
YIELD POINT 降伏点	YIELD STRENGTH AT 0.2% OFFSET 0.2% 耐力	TENSILE STRENGTH 引張強さ		ELONGATION 伸び	TEST TEMP. 試験温度	ABSORBED ENERGY 吸収エネルギー					
(-) N/mm ² ksi	(≥460) N/mm ² ksi	(≥550) N/mm ² ksi		≥19 %	(-60) °C -76 °F	AVG. 平均 (≥27) J ≥20 ft-lbf		EACH (≥21) ≥15			
WELDING CONDITIONS 溶接条件						POSTWELD HEAT TREATMENT 溶接後熱処理					
TYPE OF CURRENT 電流の種類	AC	SHIELDING GAS シールドガス									
AMPERAGE 溶接電流	115 A					(620) °C × 1 h 1148 °F × 1 h					
ARC VOLTAGE アーク電圧	21 V										
WE HEREBY CERTIFY THAT THE ABOVE TEST RESULTS AND MECHANICAL PROPERTIES SPECIFICATIONS SATISFY THE REQUIREMENTS OF THE APPLICABLE SPECIFICATION. 上記試験結果および機械的性質保証値は、適用規格の要求を 満足していることを証明します。						◆KOBEL STEEL, LTD. WELDING BUSINESS IBARAKI PLANT 株式会社 神戸製鋼所 溶接事業部門 茨城 工場  APPROVED BY CHIEF INSPECTOR 主任検査員 寺崎 清					

110318163919381819110821610482191

(注) 単位、適用規格・種類を国産大文字を使用しております。

REMARKS: CAPITAL LETTERS ARE USED EXCEPT FOR UNIT AND APPLICABLE SPECIFICATION / CLASSIFICATION.

TKW110801 100000

INSPECTION CERTIFICATE
(検 査 証 明 書)PURCHASER
需 要 家WELDING ROD OR WIRE
溶加棒又はワイヤ

CERTIFICATE No.

証明書番号: 520113299

DATE OF ISSUE

発行日 : 2011.10.28

TRADE DESIGNATION 品 名 (型 番)	DIMENSION 寸 法 (mm)	MFG.No 製 造 番 号	APPLICABLE SPECIFICATION AND CLASSIFICATION 適 用 規 格 及 び 種 類
TG-S1N	(2.4) mm 3/32 inch	GZ00820	AWS A5.28 ER70S-G ASME SFA-5.28 ER70S-G

CHEMICAL COMPOSITION 化学成分(%) ACCORDING TO EN10204 TYPE 3.1

ELEMENTS 成分	C	SI	MN	P	S	CU	NI	MO					
ROD OR WIRE 棒又はワイヤ	0.06	0.31	1.04	0.006	0.009	0.11	0.80	0.14					
ELEMENTS 成分									FS = FERRITE % (SCHAEFFLER DIAGRAM)				
ROD OR WIRE 棒又はワイヤ									FN = FERRITE NUMBER (DELONG DIAGRAM)				
									FNW = FERRITE NUMBER (WRC)				

TENSILE PROPERTIES OF DEPOSITED METAL 溶着金属の引張性能				IMPACT PROPERTIES OF DEPOSITED METAL 溶着金属の衝撃性能			HARDNESS PROPERTIES 硬さ
YIELD POINT 降 伏 点	YIELD STRENGTH AT 0.2% OFFSET 0.2% 耐力	TENSILE STRENGTH 引 張 強 さ	ELONGATION 伸 び	TEST TEMP. 試験温度	ABSORBED ENERGY 吸 収 エ ネ ル ギ ー		
(-) N/mm ² ksi	(≥360) N/mm ² ≥52 ksi	(≥480) N/mm ² ≥70 ksi	≥24 %	(-60) °C -76 °F	AVG. 平均 (≥27) J ≥20 ft-lbf	EACH (≥21) ≥15	

WELDING CONDITIONS 溶 接 条 件				POSTWELD HEAT TREATMENT 溶 接 後 熱 処 理	
TYPE OF CURRENT 電 流 の 種 類	DOEN	SHIELDING GAS シールドガス	AR 100%		
AMPERAGE 溶 接 電 流	220 A			(-) °C X - h	
ARC VOLTAGE アーク電圧	14 V			- °F X - h	

WE HEREBY CERTIFY THAT THE ABOVE TEST RESULTS
AND MECHANICAL PROPERTIES SPECIFICATIONS SATISFY
THE REQUIREMENTS OF THE APPLICABLE SPECIFICATION.

上記試験結果および機械的性質保証値は、適用規格の要求を
満足していることを証明します。

◆KOBE STEEL, LTD.

WELDING BUSINESS

FUKUCHIYAMA PLANT

株式会社 神戸製鋼所 溶接事業部門

福知山工場

APPROVED BY CHIEF INSPECTOR

主任検査員 承認済

1110261407190728191110214104R2191

(注) 単位、適用規格・種別を除き大文字を使用しております。

REMARKS: CAPITAL LETTERS ARE USED EXCEPT FOR UNIT AND APPLICABLE SPECIFICATION / CLASSIFICATION.



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Material Certificate

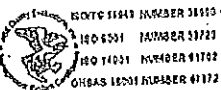


V & M VALLOUREC & MANNESMANN TUBES

V & M do BRASIL S.A.

BARREIRO PLANT - Belo Horizonte - MG - Brazil

CEP: 30161-970 - PO BOX: 1453



INSPECTION CERTIFICATE

(According to DIN EN 10204 3.1)

Nº.: 0030021384
Sheet: 1 / 2

Customer: HUPSTEEL LIMITED
Customer Order: AR2363-0311
Inspection: V & M do BRASIL S.A.

Country: Singapore
Work Order: 294024 / 000010
Material Number: 384831
Customer Material Number:

DIMENSION: 168.30mm X 7.11mm SCHEDULE: 040 # GRADE: GR 6
METALLURGICAL STANDARD: ASTM A 333 01.04.2010 # DIMENSIONAL STANDARD: ASTM A 333 01.04.2010
IN ACCORDANCE ALSO TO THE STANDARDS: ASME SA-333 / 2010 #
PRODUCT: SEAMLESS STEEL PIPE, HOT FINISHED, BEVELED ENDS 37.5 DEG. # NORMALIZED # SURFACE PROTECTION: LACQUER #
TOLERANCES: OUTSIDE DIAMETER: - 0.800mm / + 1.600mm WALL THICKNESS: - 0.889mm / + 1.600mm
CUT LENGTH: FIXED 6000.00mm
STANDARD MARKING: VMB ASTM A/ASME SA-333 6 S HF SCH 040 LT -50 C HEAT "HEAT"
SHIPPING MARKING: HUPSTEEL * P.O NO.: 5001122* SINGAPORE * NACE MR 01.75

Heat N.	Inspection Lot	Pieces
109285	030001927413	284
109286 /	030001927414	100
Total		384

THE PRODUCT IS SATISFACTORY IN THE FOLLOWING TESTS/INSPECTIONS: DIMENSIONAL # VISUAL # FLATTENING TEST # ELECTROMAGNETIC TEST ASTM E 570 # HYDROSTATIC TEST: 12.0 MPA 5 s #

Chemical Composition(%)		CEq according: C+Mn/6+(Cr+Mo+V)/5 +(Ni+Cu)/15										DI according:		Unit:		PCMI:	
		C	Mn	P	S	Si	Ni	Cr	Mo	Al	Cu	V	Nb	Ti	CEq		
Required:	Min	0.29				0.10											
	Max	0.20	1.06	0.025	0.010		0.25	0.20	0.06		0.350				0.410		
Heat N./Inspection Lot																	
109285	030001927413	0.10	0.91	0.012	0.004	0.17	0.01	0.04	0.01	0.023	0.010	0.000	0.019	0.001	0.263		
	Check1	0.11	0.90	0.012	0.004	0.18	0.01	0.04	0.01	0.026	0.006	0.000	0.018	0.001	0.269		
	Check2	0.11	0.93	0.013	0.006	0.18	0.01	0.05	0.01	0.023	0.006	0.003	0.020	0.001	0.278		
109286	030001927414	0.10	0.94	0.014	0.005	0.16	0.01	0.05	0.01	0.025	0.000	0.000	0.019	0.001	0.269		
	Check1	0.11	0.94	0.015	0.005	0.17	0.01	0.05	0.01	0.028	0.003	0.000	0.020	0.001	0.276		
	Check2	0.11	0.95	0.015	0.004	0.17	0.01	0.05	0.01	0.025	0.004	0.000	0.019	0.001	0.276		

Chang
LOCAL CHINA
Chang
Inspection QA/QC Division
Integrity and Reliability
Chang

Mechanical Tests S.Direction: Longitudinal Temperature: Room Temperature Gage Length: L0=2"

		Area(MM2)	YS(MPA)	TS(MPA)	E(%)
Required:	Min		240	415	28
	Max				
Heat N./Inspection Lot					
109285	030001927413	177.4	331	441	40
109286	030001927414	192.2	329	437	42

YS-Yield Strength ; TS-Tensile Strength ; E-Elongation

CERTIFIED TRUE COPY

BY DELTAMETAL (THAILAND) LIMITED
P.O.NO. 145015361
ITEM NO. 096085



VALLOUREC & MANNESMANN TUBES



ISO 9001 NUMBER 31220
REG 0031 JARASER 03723
ISO 14001 NUMBER 11702
GHSA 11001 NUMBER 47872

INSPECTION CERTIFICATE

(According to DIN EN 10204 3.1)

Nº.:
0030021384

Sheet:
2 / 2

Hardness

		HV
Required:	Min	
	Max	246
Heat N./Inspection Lot		
109285	030001927413	145
109286	030001927414	146

Impact According: CHARPY 10X55X5 V NOTCH Direction: Longitudinal Temperature: -50 °C
AE1 (J) AE2 (J) AE3 (J) AE Aver. (J)

Required:	Min	14	14	14	17.0
	Max				
Heat N./Inspection Lot					
109285	030001927413	98	110	114	107.3
109286	030001927414	122	120	116	119.3
AE-Absorbed Energy					

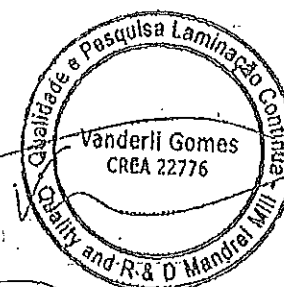
Remarks:

- MATERIAL IN ACCORDANCE TO ITEM A.2 NACE MR 0175/ISO 15156-2/2003.

- FINAL HOT ROLLING OPERATION FINISHED ABOVE 850°C AND COOLED IN STILL AIR

We would like to confirm that for the above stated item, the pipes were stencilled with customer name as #HUPSTBL# instead of #HUPSTBEL# due to the problem in our computer system.


Qualidade e Pesquisa Laminacao Controllada
Vanderli Gomes
CREA 22776
Inspection QA/QC Division
Integrity and Reliability
Chambers



We hereby certify that this product has been manufactured and examined in accordance with all requirements of the standards and specifications and all the results are found to be satisfactory. This testimonial and certificate respectively is recorded by a computer system and is valid without signature. Alteration or use for others products are regarded as falsification of documents and will be subject to criminal jurisdiction.

QUALITY CONTROL DEPARTMENT

FAX: (55-31) 3328-2587

e-mail: CONT.QUALIDADE@VMTUBES.COM.BR

DATE
30.05.2011

LUIZ ALEXANDRE COSTA - CREA/MG 69.543/D
TECHNICAL RESPONSIBLE

Eco Tubes: The tubes from V&M do BRASIL are manufactured with steel which uses charcoal as a source of energy in its production. This coal comes from more than 100.000 ha of forest planted by V&M Florestal. With the acquisition of 67,9 ton(s) of steel tubes from V & M do BRASIL, your company contributed to the reduction of the greenhouse effect, avoiding the accumulation of 122,2 ton(s) of Carbon Dioxide CO2 in the atmosphere.

CERTIFIED TRUE COPY

BY DELTA METAL (THAILAND) LIMITED

P.O. NO. 46019261
FROM NO. 2000